

Books on Computational Physics

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- Ilya Prigogine and Stuart A. Rice, New Methods in Computational Quantum Mechanics, Wiley (1997).
- [Dennis Rapaport](#), The Art of Molecular Dynamics Simulation, Cambridge University Press (1996). The text is partly tutorial, but also contains many computer programs for practical use. The book will be of value to advanced students and researchers in physics, chemistry, polymer science and materials science. Copies of the C program listings are available at <http://www.cup.cam.ac.uk/onlinepubs/ArtMolecular/ArtMoleculartop.html>. [Table of Contents](#).
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- Dietrich Stauffer, ed., *Annual Reviews of Computational Physics 3* (1995). Articles on "Computer Simulations of Supercooled Liquids and Structural Glasses" (W. Kob), "Evolution Motivated Computer Models" (N. Vandewalle and M. Ausloos), "A Survey of Genetic Algorithms" (M. Tomassini), "Parallel Algorithms for Short-Range Molecular Dynamics" (D. M. Beazley et al.), "Dynamics of Nonequilibrium Deposition with Diffusional Relaxation" (V. Privman), "Recent Advances in Lattice Boltzmann Computing" (Y. H. Qian et al.), and "Condensed Matter Physics at One Tera-Kelvin" (J. Potvin).
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Theory Applied to Concrete" (E. J. Garbocz and D. P. Bentz), "Computer Simulations in Urban Geography" (L. Benguigui), "Large-Scale Traffic Simulation for Transportation Planning" (K. Nagel, J. Esser and M. Rickert), "Biological Evolution Through Mutation, Selection, and Drift: An Introductory Review" (E. Baake and W. Gabriel), "An Evolutionary Model for Simple Ecosystems" (F. Bagnoli and M. Bezzi), "Microscopic Simulation for Reaction-Diffusion Processes," and "Applications of Population Biology and Product Marketing" (E. Bettelheim and B. Lehmann).

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Updated 8 December 2022.